

A COLLABORATION BETWEEN
ANDERSON SERANGOON JUNIOR COLLEGE &
NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 3

CANDIDATE
NAME

CENTRE
NUMBER

CHEMISTRY

9813/01

Paper 1

22 September 2020

2 hours 30 minutes

Candidates answer on the Question Paper.

Additional Materials: Data Booklet
 Insert

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer all questions in the spaces provided on the Question Paper. If additional space is required, you should use the pages at the end of this booklet. The question number must be clearly shown.

Section A

Answer **all** questions.

Section B

Answer **two** questions.

The use of an approved scientific calculator is expected, where appropriate.
A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Setters

Q1: NYJC

Q2: ASRJC

Q3: ASRJC

Q4: ASRJC

Q5: ASRJC

Q6: NYJC

Q7: NYJC

This document consists of **50** printed pages and **1 insert**

Section A
Answer all questions in this section.

Q1 Setter: NYJC

- 1 The information provided in the insert is taken from a published scientific article. Other published articles may not agree with all of this information.

You should read the whole insert before your start to answer any questions and use the information it contains to answer the questions.

- (a) Suggest a reason why CO₂ application to organic reactions are quite limited. [1]

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- (b) In Abstract 1, the experiment was carried out in a 40 cm³ reaction vessel with 0.34 MPa of carbon dioxide at 30 °C.

- (i) Calculate the mass of *p*-terphenyl used in the reaction. [2]

- (ii) Suggest the role of *p*-terphenyl and give two evidences to support your answer. [2]

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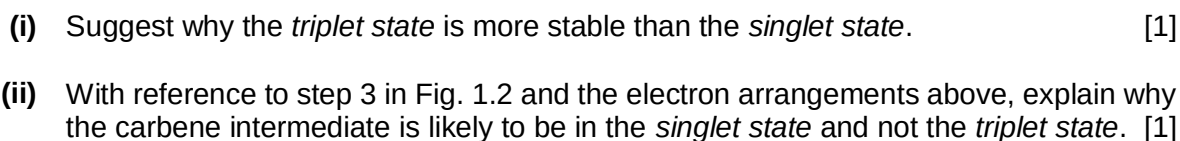
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- (c) In the research laboratory, the saturated calomel electrode (SCE) is used in place of the standard hydrogen electrode (SHE). The SCE is a reference electrode based on the reaction between elemental mercury and mercury(I) chloride. The standard electrode potential of SCE with respect to the SHE is shown below.



- (i) Calculate the E (p -terphenyl / p -terphenyl $^{-\bullet}$) and E (CO_2 / $\text{CO}_2^{-\bullet}$) with respect to SHE in Abstract 1. [1]
- (ii) Hence or otherwise, calculate the E_{cell} and standard Gibbs free energy change, ΔG , of step 3 in Fig. 1.1. [2]
- (iii) Draw the dot-and-cross diagram of $\text{CO}_2^{-\bullet}$ and suggest its bond angle. [2]

[illegible]

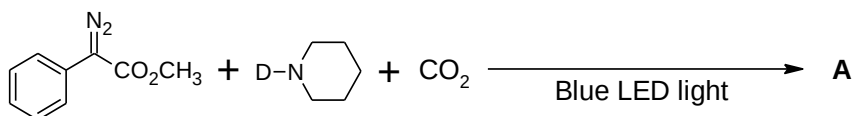


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- (e) (i) Suggest the type of reaction occurring in step 4 of Fig. 1.2 and deduce the role of CO_2 . [2]
- (ii) Suggest the mechanism in step 5 of Fig. 1.2. [1]

[illegible]

- (f) The reaction described in Abstract 2 can also obtain a carbamate in the absence of tetrahydrofuran (THF). An isotope labelling experiment was carried out using deuterated amine shown below. Suggest the structure of compound **A**, clearly showing the position of the deuterium. [1]



- (g)** In both Abstract 1 and 2, light is needed for such photochemical reactions to occur. In addition, both reactions can only take place with specific substituents on the reactants.

- (i) With reference to Fig. 1.1, explain briefly why the reaction described in Abstract 1 is a photoredox reaction. [1]

- (ii)** Suggest why red LED light is not commonly used in photochemical reactions. [1]

- (iii) By considering the stability of reaction intermediates in Fig. 1.1 and 1.2, suggest why R_1 and R_2 must be aromatic rings. [2]

[Turn Over

Q2 Setter: ASRJC

2 (a) (i) Draw the molecular orbital diagram of F_2^+ and hence, use it to calculate the bond order of the ion. [3]

(ii) Identify the LUMO and HOMO of F_2^+ . [1]

(iii) A substance that consists of one or more unpaired electrons displays paramagnetism while a substance with no unpaired electron displays diamagnetism.

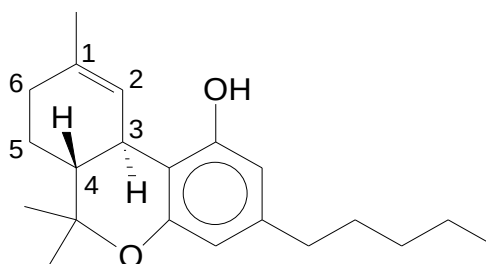
Suggest, with a reason, whether F_2^+ is paramagnetic or diamagnetic. [1]

[illegible]

- (b)** Using Linear Combination of Atomic Orbitals (LCAO), draw the shape of the bonding and anti-bonding molecular orbitals formed by the 2p orbitals of two adjacent unsaturated carbon atoms in penta-1,3-diene. [2]

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- (c) Tetrahydrocannabinol, THC, is a major constituent of cannabis. One enantiomer of THC is (R,R) Δ^1 – tetrahydrocannabinol (THC).



(R,R) Δ^1 – tetrahydrocannabinol (THC)

Carbon atoms 3 and 4 are chiral centres.

- (i) Explain the underlying principles of UV/VIS spectroscopy. Details of instrumentation are not required. [2]
- (ii) Identify the chromophore in the THC molecule. [1]
- (iii) Suggest whether the UV spectrum of Δ^2 – THC (i.e. the isomer with the double bond between carbon atoms 2 and 3) will be different from that of Δ^1 – THC. Explain your answer. [1]
- (iv) An intravenous drip solution of the enantiomer of THC, (S,S) Δ^1 – tetrahydrocannabinol, is used in a clinical trial to treat high blood pressure. Serial dilution of the drip solution was carried out:
 - 1 cm³ of the drip solution was diluted with 4 cm³ of water.
 - 1 cm³ of the resultant solution was then diluted with 4 cm³ of water.

This diluted drip solution shows absorbance of 0.954 at a wavelength of 279 nm in a cell with a path length of 1.0 cm. The enantiomer of THC, (S,S) Δ^1 – tetrahydrocannabinol, has a molar extinction coefficient of 2010 dm³ mol⁻¹ cm⁻¹.

Calculate the concentration of the enantiomer in the drip solution. [2]

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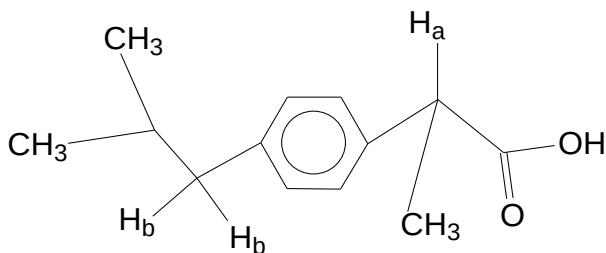
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[Turn Over

3 (a) Ibuprofen contains 18 protons in various chemical environments. Two types of protons are described as H_a and H_b in the structure below.



- (i)** Outline the basic principles of ^1H NMR. [3]
- (ii)** Explain the use of TMS in ^1H NMR spectroscopy. [1]
- (iii)** Copy the above structure of ibuprofen, identifying the different proton by appropriate symbols such as H_c , H_d [1]
- (iv)** Suggest with reasons the splitting patterns and approximate chemical shift values of the H_a and H_b protons in the ^1H NMR spectrum. [2]

[illegible]

- (b) Compound **E** contains the elements carbon, hydrogen, nitrogen and oxygen. Compound **E** reacts with aqueous bromine in a mole ratio of 1:3.

The mass spectrum of Compound **E** is shown in Fig. 3.1. The relative intensity of the molecular ion at m/z 165 is 29.9% and that of ($M+1$) ion at m/z 166 is 2.96%.

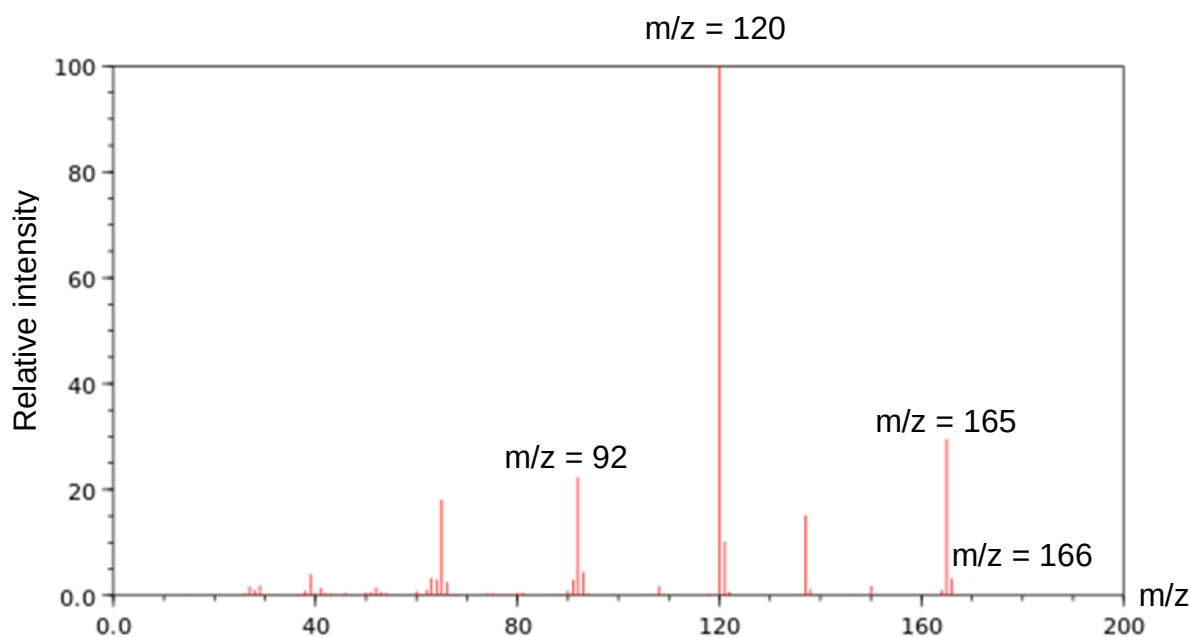


Fig. 3.1

The infra-red (IR) spectrum of Compound **E** is shown in Fig. 3.2.

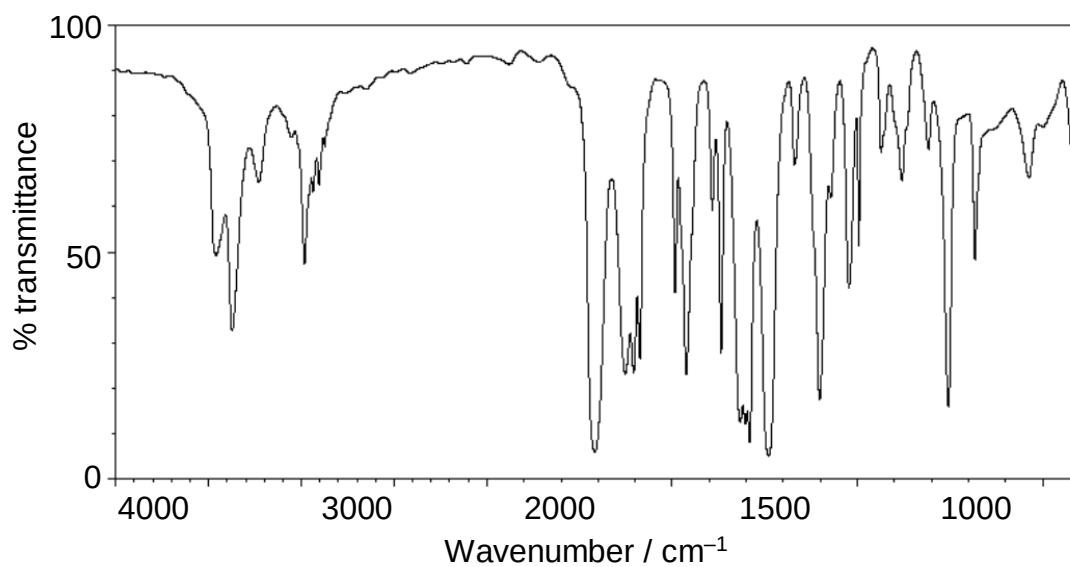


Fig. 3.2

The NMR spectrum of Compound **E** shows the following signals in Table 3.3.

Table 3.3

δ / ppm	splitting pattern	number of protons
1.40	triplet	3
3.80	singlet	2
4.30	quartet	2
6.84 – 7.42	multiplet	4

The signal at δ 3.80 disappears when D₂O was added.

Use the above data to deduce the structure of **E**. Explain your reasoning fully.

[9]

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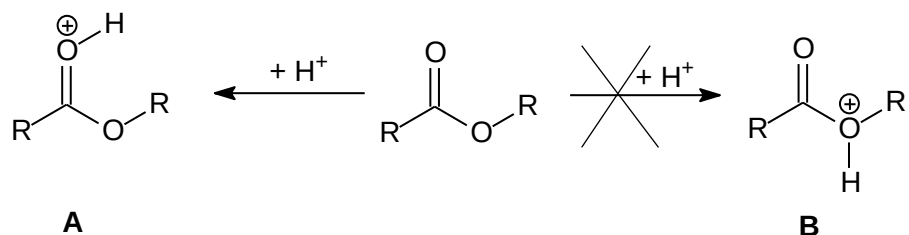
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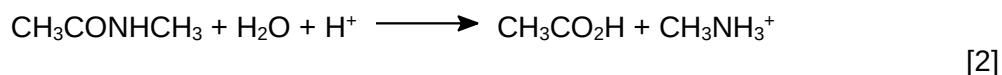
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[Turn Over

For esters, the protonation that occurs during the first step produces the more stable cation **A** rather than cation **B**.



- (i) Suggest why cation **A** is more stable than cation **B**. [1]
- (ii) Use the information given above to suggest a mechanism for the following reaction.



- (iii) Using your mechanism in (a)(ii), suggest why amides undergo hydrolysis at a much slower rate than ester. [2]

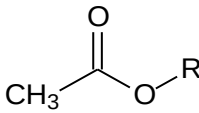
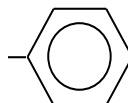
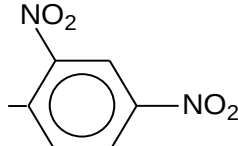
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- (b) Ester undergoes base-catalysed hydrolysis. It follows the rate equation shown.

$$\text{rate} = k[\text{OH}^-][\text{ester}]$$

Experiments were carried out to investigate the hydrolysis of some esters at high pH. The results are shown in Table 4.1.

Table 4.1

ester	-R in 	$k / \text{mol}^{-1} \text{dm}^3 \text{s}^{-1}$
P	-CH ₂ CH ₃	0.11
Q	-C(CH ₃) ₃	1.5×10^{-3}
R		1.4
S		to be predicted

- (i) Explain why the value of k for ester **Q** is smaller than that of ester **P**. [1]
- (ii) Suggest if the value of k for ester **S** will be larger or smaller than that of ester **R**. Explain your answer. [2]

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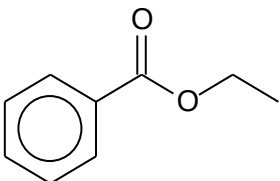
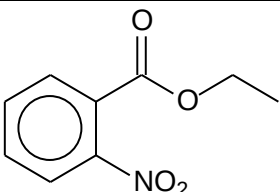
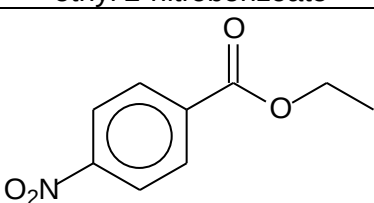
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- (c) The hydrolysis of three esters using aqueous NaOH is studied. Table 4.2 shows the activation energies and enthalpy changes for the three esters.

Table 4.2

Ester	$E_a / \text{kJ mol}^{-1}$	$\Delta H / \text{kJ mol}^{-1}$
 ethyl benzoate	+55.4	+52.8
 ethyl 2-nitrobenzoate	+48.6	+46.0
 ethyl 4-nitrobenzoate	to be calculated	+45.8

The Bell-Evans-Polanyi Principle relates the activation energies of a series of related reactions to their enthalpy changes.

- (i) Using Table 4.2, calculate a value for the activation energy for the base-catalysed hydrolysis for ethyl 4-nitrobenzoate. [2]
- (ii) Draw the structure of the transition state for the rate determining step in the base-catalysed hydrolysis of ethyl benzoate. [1]

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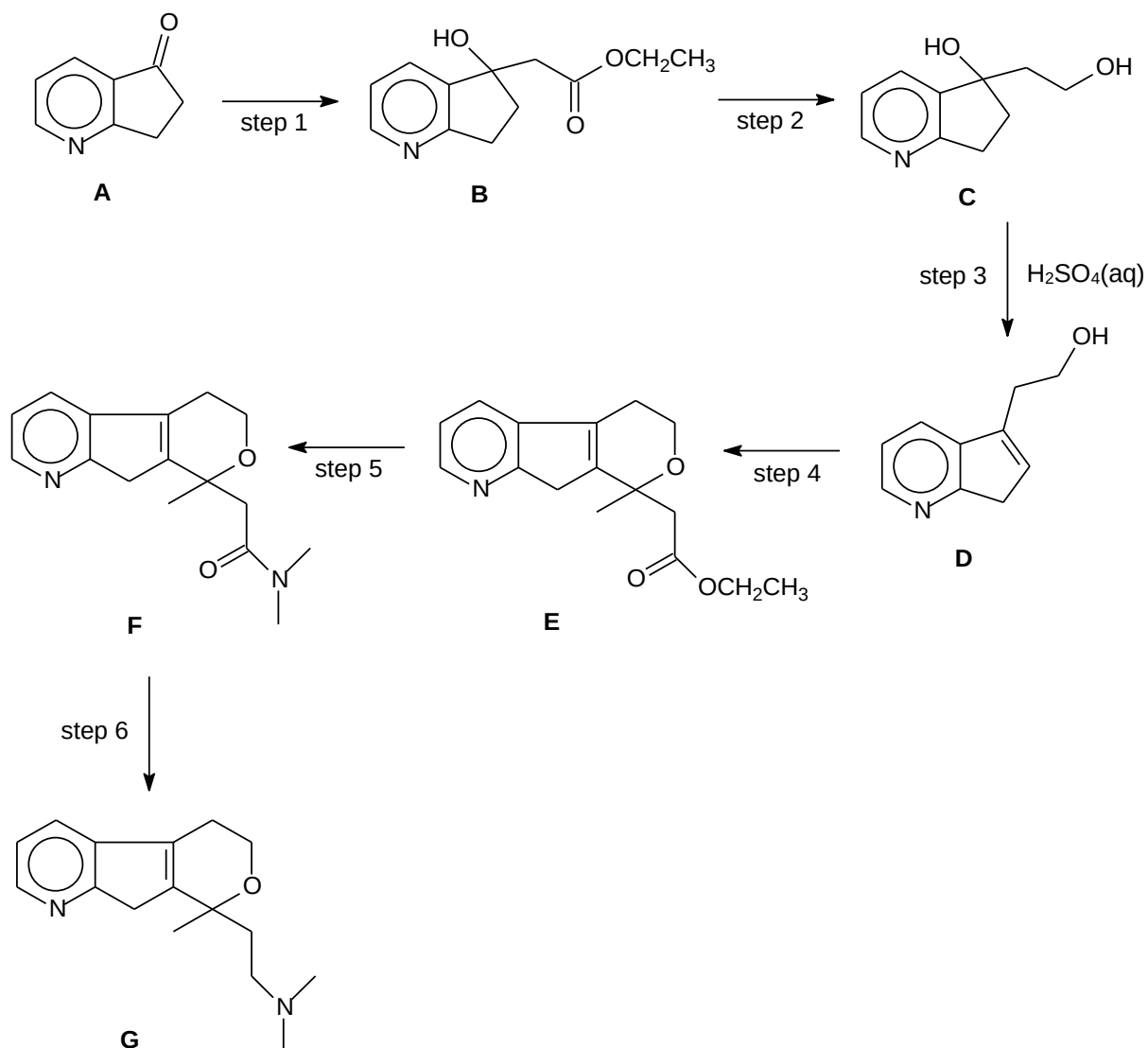
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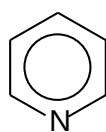
H3 Chemistry 9813/01 NYJC J2/2021 PX

Section B

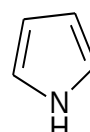
Answer **two** questions in this section.**Q5 Setter: ASRJC****5 (a)** **G** can be synthesised according to the following reaction scheme in Fig. 5.1.**Fig. 5.1**

- (i) Explain how you would confirm the complete conversion of **A** to **B** in step 1 using a simple chemical test. [2]
- (ii) Explain one difference in infra-red absorption frequency between **A** and **B**. [1]
- (iii) Suggest a single reagent which could be used in both step 2 and step 6. [1]
- (iv) Given that step 3 has an overall first order kinetics, name and draw the reaction mechanism for the conversion of **C** to **D** in step 3. [3]

- (b)** Both pyridine and pyrrole are aromatic as it has $[4n+2]\pi$ electrons delocalised in a ring (where $n=1, 2, 3, \dots$). Both also undergo the same type of substitution reaction as benzene.

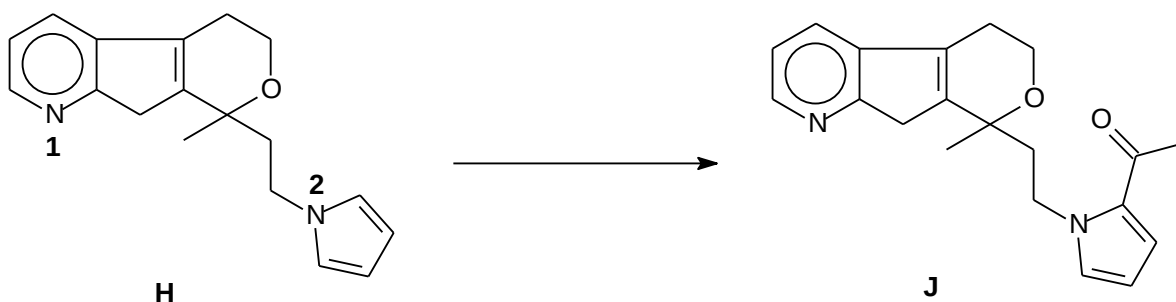


pyridine



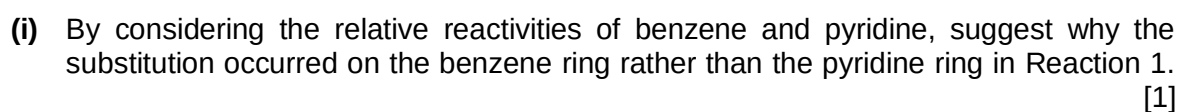
pyrrole

Compound **H** is an analogue of **G** and it consists of a pyridine group and a pyrrole group.



- (i) State the hybridisation of nitrogen in N1 and N2. [1]
- (ii) Explain why N1 is less basic than a secondary amine. [1]
- (iii) Explain why N2 is less basic than N1. [1]

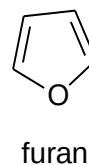
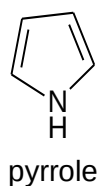
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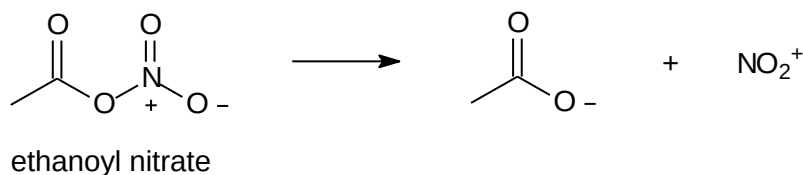
(ii) Suggest the mechanism for the reaction between **L** and CH_3ONa in Reaction 2. [2]

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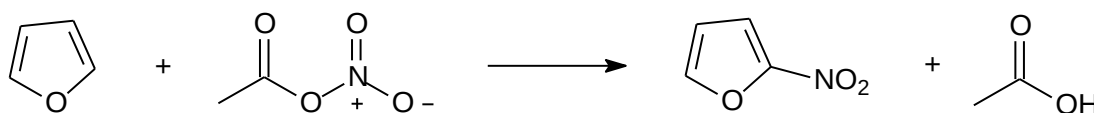
- (d) Like pyrrole, furan is a five-membered heterocyclic compound that can be found in a variety of medicines and is used in the synthesis of active drug molecules.



Furan is readily nitrated to 2-nitrofuran using ethanoyl nitrate which generates the electrophile NO_2^+ .



The overall equation for the nitration of furan is shown below.



- (i) Complete Fig 5.2 to suggest the mechanism for this reaction. [2]

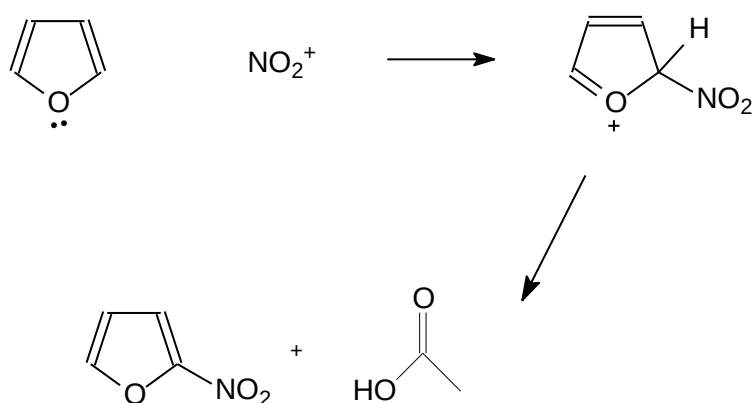


Fig 5.2

- (ii) 2-nitrofuran is the major product and 3-nitrofuran is the minor product in the reaction above. With the aid of suitable diagrams, explain this outcome by considering the stability of the cationic intermediates involved. [3]
- (iii) Predict the relative rate of nitration of pyrrole compared to furan. Explain your reasoning. [2]

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(a)

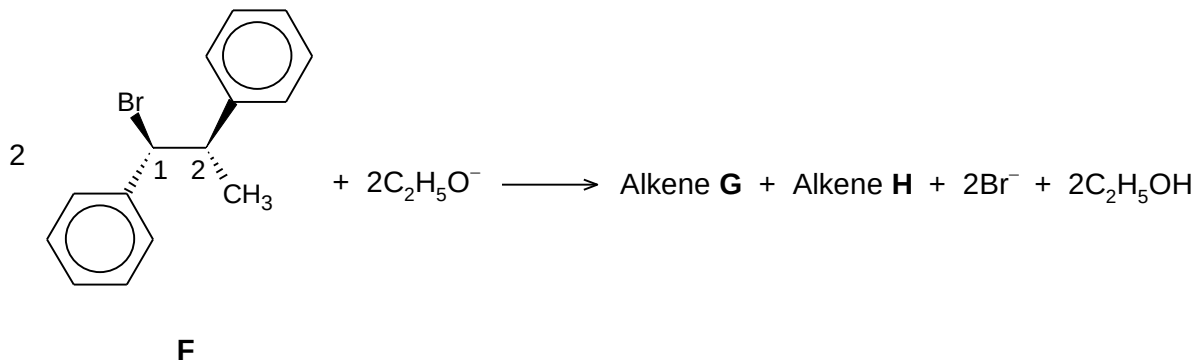
(ii)

[3]

(iii)

[illegible]

- (b) When heated with ethoxide, $\text{C}_2\text{H}_5\text{O}^-$, compound **F** undergoes E2 elimination to form alkenes **G** and **H**. Alkene **G** is formed via anti periplanar arrangement whereas alkene **H** is formed via syn periplanar arrangement.



- (i) Assign the stereochemistry (*R* or *S*) at each of the carbon atoms 1 and 2 in **F**, and explain your answer. [2]
- (ii) Draw Newman projections along the C1–C2 bond for **F** to illustrate the conformations that are required for elimination to occur to produce alkenes **G** and **H** respectively. Label your diagrams appropriately. [2]
- (iii) Suggest the structure and assign their stereochemistry (*E* or *Z*) of alkenes **G** and **H**. [2]
- (iv) Hence, predict whether alkene **G** or **H** has a higher yield. Explain your answer in terms of the stability of the conformations in (b)(ii). [2]
- (v) Draw suggested mechanism for the formation of alkene **G**. [1]

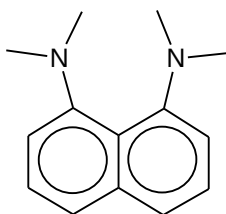
[illegible]



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- J**

[illegible]

- (d) Chlorocyclohexane reacts with aqueous NaOH to form both cyclohexanol and cyclohexene whereas it reacts with compound **L** ($pK_a = 12.3$) to form cyclohexene as the only product.

**L**

Comment on the difference in the observations.

[2]

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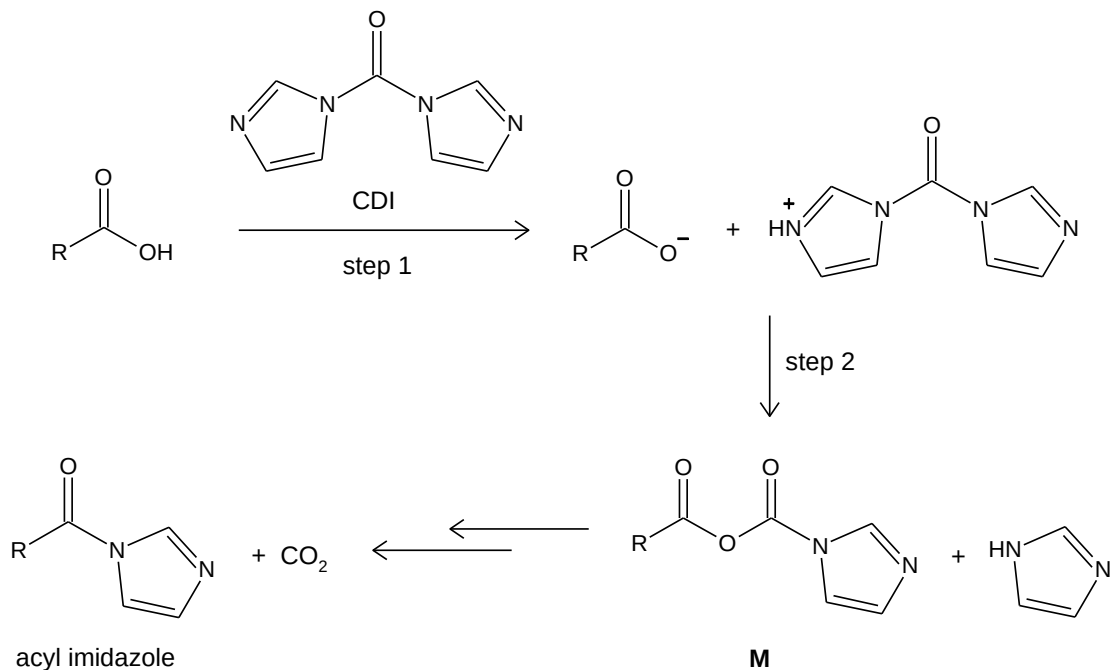
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Q7 Setter: NYJC

- 7 Acyl imidazoles have recently been used to obtain structural information of a particular RNA through the reaction with the 2'-hydroxy group of ribose ring. They are also useful in the synthesis of amides as they are still functional in aqueous conditions.

Acyl imidazoles can be synthesised by reacting a carboxylic acid with carbonyldiimidazole (CDI) via a multi-step mechanism.

- The first step is an acid-base reaction between the carboxylic acid and CDI.
- The products from the first step will then react to form intermediate **M**.
- Intermediate **M** will then form another intermediate with a four and five membered rings which subsequently forms the acyl imidazole.



- (a) (i) Draw suggested mechanism for the synthesis of acyl imidazole from step 2 onwards. [3]
- (ii) Carboxylic acids are usually converted to acyl chlorides to synthesise amides.

Suggest why acyl chlorides are less preferred than acyl imidazoles for the synthesis of amides. [1]

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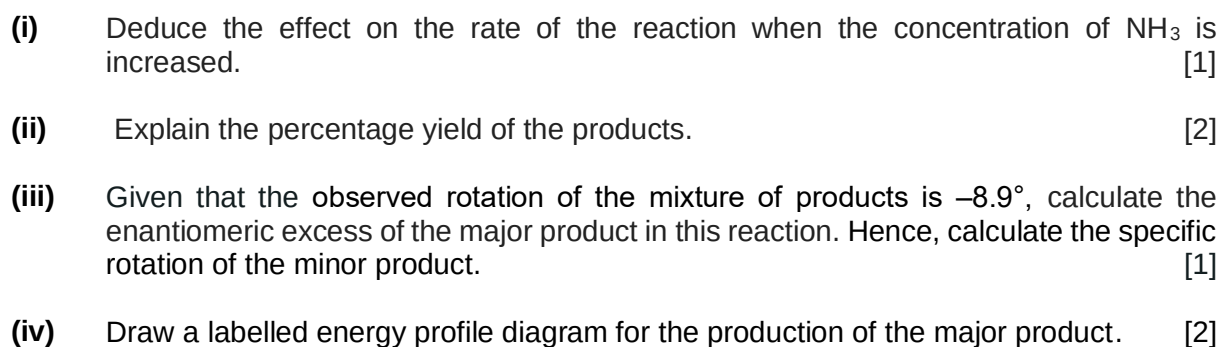
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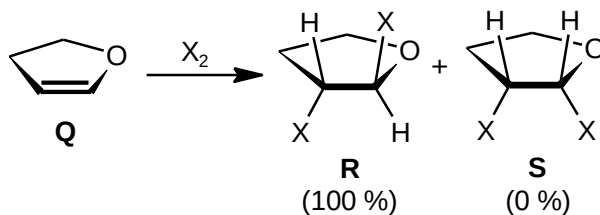
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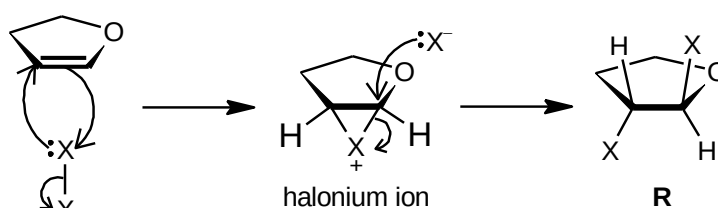
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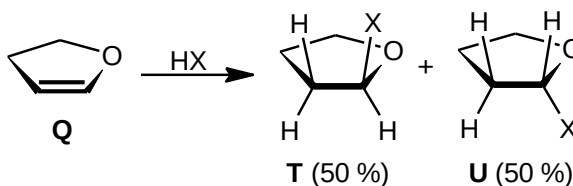
- (c) In an electrophilic addition, a compound with a double bond reacts with an electrophilic reagent to produce an intermediate that subsequently reacts with a nucleophile. Compound **Q** undergoes electrophilic addition with liquid halogen, X_2 , to give solely compound **R**.



To explain the stereospecificity, George Kimball and Irving Roberts proposed that the reaction intermediate is not a carbocation but is instead a halonium ion formed by electrophilic addition of halogen to the alkene. Subsequently, the halide ion attacks the halonium ion to give compound **R**.



However, when the reaction is repeated with hydrogen halide, no such stereospecificity is observed.



- (i) State the stereochemical relationship between **R** and **S**. [1]
- (ii) Suggest why **S** is not formed. [1]
- (iii) Suggest why no stereospecificity is observed but a racemic mixture is formed when hydrogen halide is used. [2]

(*R*)-3-methylcyclohexene reacts with W_2 and VW under different conditions to give **Y** and **Z** respectively. **V** and **W** are Group 17 elements. Some mass spectra details are given in Fig. 7.1.

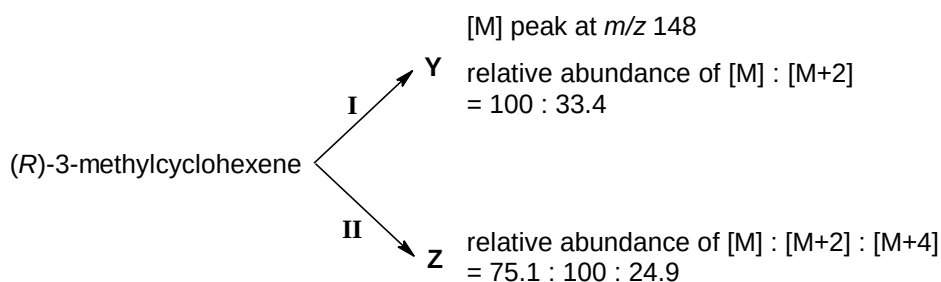


Fig. 7.1

- (iv) Deduce the identity of **V** and **W**. [2]
- (v) Deduce the molecular formula of **Y**. Suggest the reagents and conditions for reaction I. [2]
- (vi) Suggest the structures of **Y** and **Z**, showing clearly the stereochemistry. [2]

[illegible]

[Turn Over

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